

SMART TRANSIT NETWORK SYSTEM

Work Distribution Among Group Members

Topic 11 | Group 803

Group Members

Student ID	Name	Primary Responsibility
23201452	Alma Usha	Static Routing
23201158	Maisha Jahan Fatima	Email Services
23201464	Abu Bakar Siddique	Dynamic Routing
22299398	Samiul Mahmud	DHCP

1. Alma Usha – Static Routing (Primary)

Configured all five required static route types across the network: the next-hop static route on R-CCH (to the FDU network, using the DPC interface IP as next-hop), the exit-interface static route on R-RSP (to the CCH network, via the FDU-facing serial interface), the recursive static route on R-DAU (to the RSP network, with next-hop PSC) together with the corresponding recursive route on R-PSC (to RSP, via CCH), the floating static route between R-DPC and R-PSC over the backup serial link (administrative distance 130), and the default static route on R-RSP pointing to FDU. Also verified that no summary or fully-specified static routes were used anywhere in the design.

2. Maisha Jahan Fatima – Email Services (Primary)

Set up the SMTP/POP3 email servers on CCH and PSC, created user accounts for every centre, and configured the mail clients on all PCs so that centres without a local mail server (DPC, FDU,

RSP, DAU) relay their outgoing and incoming mail through the PSC server. Verified successful send/receive between accounts on the same domain and documented the platform limitation that prevents true multi-hop SMTP relay between the two separate mail servers (CCH and PSC) in Cisco Packet Tracer.

3. Abu Bakar Siddique – Dynamic Routing (Primary)

Configured RIPv2 on CCH, DPC, FDU and PSC (and confirmed RIP was left disabled on RSP and DAU, as required), configured redistribution of CCH's static routes into the RIP domain, and applied passive-interface on the DPC–PSC backup serial link on both ends so that the backup link is used only by the floating static route and never by RIP. Also carried out the end-to-end verification: failover testing (shutting down the CCH–DPC link and confirming the floating route activates automatically and recovers correctly), and connectivity testing (ping/traceroute) across every required route.

4. Samiul Mahmud – DHCP (Primary)

Derived the base network address from his student ID (22299398 → sum of last three digits $3+9+8 = 20$ → base network 20.20.0.0/16) and performed the VLSM subnetting for all six LAN subnets and six point-to-point WAN links. Configured router-based DHCP pools on DPC and FDU (with appropriate excluded address ranges, default gateway, DNS server and domain name), and configured the dedicated DHCP server on the PSC LAN, sizing the address pool to comfortably cover PSC's 135 devices.

Shared and Secondary Responsibilities

Task	Owner
Topology design, device placement/cabling in Packet Tracer, subnet labels/notes	Alma Usha
DNS server (CCH) and Web server (PSC) configuration	Maisha Jahan Fatima
Router naming/security, interface addressing for statically-addressed hosts	Abu Bakar Siddique
IP addressing table and VLSM tree	Samiul Mahmud
Verification screenshots, testing log, and final report compilation	All members (joint)